



PROFESSIONAL SUMMARY

Staff Software Engineer with **9+ years of total software engineering experience** (including 5.5+ years at Visa Inc.) architecting and delivering cloud-native platforms for real-time payment fraud detection at global scale. Extensive experience designing cloud data ingestion REST APIs, building Kubernetes infrastructure (Helm charts, ENV0/Terraform IaC), engineering GenAI-powered tooling, and owning end-to-end system architecture across AWS. Contributor to a **USPTO-granted patent** (Release Management Automation). Proficient Gen AI practitioner: daily user of **Claude, Cursor, and Cline** for full-stack development, infrastructure scripting, architecture design reviews, and technical documentation. Core technical strength in **Java 17, Python, Go**, Kafka, AWS (15+ services), and high-throughput distributed systems. Certified by **Caltech, UT Austin, Stanford, and AWS**.

TECHNICAL SKILLS

- Languages:** Java 17, Python, Go, Scala, JavaScript, TypeScript

Gen AI & ML: Claude, Cursor, Cline, MCP Servers, AI-assisted development (Go, Python, shell, Helm), architecture design reviews, technical documentation, AIP REST inference, OPA/Rego, EMR retraining

Cloud (AWS): EC2 (Nitro Enclaves), EKS, Lambda, EMR, MSK (Kafka), ElastiCache, AuroraDB, S3, API Gateway, KMS, CloudWatch, CodeBuild/Pipeline

Testing Frameworks: JMeter, Blazemeter, Selenium, REST Assured, Cypress, TestNG, Cucumber, JUnit, Postman

DevOps & Infra: Kubernetes (EKS), Helm, Terraform / ENV0 (IaC), Jenkins, CodeBuild/Pipeline, Artifactory, Git (GitHub Enterprise)
- Databases:** MySQL, PostgreSQL, Cassandra, Redis, Valkey, Neo4j, MongoDB, HDFS / Delta Lake

Performance Testing: Blazemeter (built from scratch), JMeter, RedButton (Active-Active failover)

Security: mTLS, HMAC, AES-GCM (VisaKP), PGP, AWS KMS, Nitro Enclaves, SSL/TLS, OPA/Rego, Vault

Monitoring: CloudWatch, Grafana, Prometheus, Splunk, Humio

Messaging & Streaming: Apache Kafka (MSK), MirrorMaker2, Kinesis Firehose, SNS, Spring Boot, Apache Spark

Methodologies: Agile/Scrum, Shift-Left Testing, CI/CD, Active-Active Architecture, ITDR

PROFESSIONAL EXPERIENCE

Staff Software Engineer

Visa Inc. | Austin, TX (5 yrs 9 mos)

Promoted progressively: Staff Software Engineer (Aug 2024 – Present) | Staff SDET (Apr 2023 – Aug 2024) | Sr. SDET (Nov 2020 – Apr 2023).

Engineered cloud-native platform services, REST APIs, IaC automation, and led technical delivery.

November 2020 – Present

Cloud-Native Fraud Scoring Platform — AWS (Global Payments)

Aug 2024 — Present

Engineering lead for a dual-region Active-Active real-time fraud scoring system handling UK, Brazil, Argentina, and Costa Rica payment corridors across three AWS regions — Ireland, London, and São Paulo.

Cloud Engineering & API Design

- ▶ Authored **Component Operations Map** and led technical design of Cloud Data Ingestion REST API (replacing SFTP-based batch processing); drove cross-team architecture decisions with VADP, AIP, Security, and IRP teams; eliminated 24-hour data lag for onboarding BCR (Costa Rica) and New Zealand clients
- ▶ Built **Config Stack IaC environment from scratch** (VMCS/RPCS/VEPS/RAD): authored Helm charts for all Kubernetes services, configured AWS Secrets Manager entries, executed ENV0/Terraform IaC in correct dependency sequence (IAM → S3 → ECR → EKS → ElastiCache → MSK → CodeBuild → CodePipeline)
- ▶ Engineered and validated full rule lifecycle: VRM UI → BFF (rt-api-rulesmanager) → RPCS/VMCS → RAD compiles OPA/Rego → ElastiCache Redis → VADP Decision Service → rulesMatched[] in real-time scoring response; served as **technical lead** for CTE and PROD sign-offs
- ▶ Led **iFAST client onboarding** end-to-end (QA → CTE → PROD): diagnosed HMAC token errors, resolved TLS certificate issues across 4 environments, validated mTLS handshake and API rate-limit configurations

Gen AI Engineering

- ▶ ⚡ Daily user of **Cursor and Cline** for 2+ years: accelerated **full-stack development** (Go, Java, Python, shell, Scala) through AI-assisted script authoring, Helm/Terraform/ENV0 module generation, and rapid navigation of 20+ service repositories to conduct root cause analysis.
- ▶ ⚡ Used **Claude** extensively (1+ year) as a core **development and design tool**: conducting **architecture and API design reviews** (REST contracts, data pipeline schemas, system component design); generating **production-ready** Go/Python service code and Kubernetes configurations; and authoring technical documentation (runbooks, architecture diagrams) or test plans for complex cloud scenarios.
- ▶ ⚡ Introduced **Claude Code** to A2A Champions team (TLRTPCC-2547): configured **MCP server integrations** (Jira, Jenkins, Wiki) to enable AI-driven development workflows and pipeline automation; built a **GenAI performance results analyzer** (Go/Python) that parses JMeter/Blazemeter output to generate actionable latency insights.

Performance Engineering & Infrastructure

- ▶ Built **Blazemeter performance testing infrastructure from zero** for all environments and AWS regions; established SLAs (500 TPS sustained, 1,000 burst, P99.95 ≤ 250ms); executed 8+ campaigns — **6M+ transactions/session, zero errors, P99.95 = 104.96ms**
- ▶ Identified **Lambda Authorizer cold-start latency spikes** (7–100ms+) as production regression root cause; drove architectural change moving HMAC auth into Nitro Enclave, permanently reducing API Gateway latency
- ▶ Validated A2A Gateway (Go, Nitro Enclave), VADP Decision Service (Go, EKS, gRPC), MSK Kafka pipelines, ElastiCache → Valkey migration, ITDR readiness (MirrorMaker2, cross-region failover, GTM), and zero PII leakage across all CloudWatch log streams
- ▶ Validated all five Kafka data ingestion pipelines (TransactionStatus, Fraud, AccountDetails, Entitlements, HistoricalData) — V1/V2 detection, AES-GCM PII encryption, account concatenation, KMS key management, and PAN-Data 3-day flow (SFTP → PGP decrypt → Nitro Enclave)
- ▶ Mentored **full QA team** (~1 year): Blazemeter, CloudWatch/Grafana, Active-Active test strategy, EMR Lambda validation, release sign-off; led formal KT sessions ensuring team independence

Stack: Java 17, Python, Go, Scala, AWS (EKS, EC2 Nitro Enclaves, MSK, Lambda, EMR, ElastiCache/Valkey, AuroraDB, S3, API Gateway, KMS, CloudWatch, CodeBuild), Kubernetes, Helm, Terraform/ENV0, OPA/Rego, Blazemeter, JMeter, Redis, gRPC, Spring Boot, Apache Kafka, MirrorMaker2, Kinesis Firehose, Claude, Cursor, Cline

On-Premise Real-Time Payment Risk Platform

Apr 2023 — Aug 2024

Staff SDET on Visa's on-premise real-time fraud scoring system — sub-250ms latency, Active-Active dual datacenter architecture supporting UK and Argentina payment corridors.

- ▶ Engineered Java 17 and Python service components; executed **sustained 1,600 TPS performance tests** over 3 hours — **16.9M transactions, P99.95 = 36ms, zero errors**; ran 12+ performance campaigns at 300–1,600 TPS with zero 5xx errors
- ▶ Validated end-to-end **VisaKP cryptographic migration** (legacy encryption → one-way HMAC) across Java ALR Consumer (in-place Redis re-encryption), Fraud Consumer (MySQL rebuild), and online Decision Service DAG; confirmed HMAC key determinism for correct GFL/ALR scoring
- ▶ Led **SSL/TLS weak cipher remediation** across 3 Java/REST APIs; validated Cassandra TLS upgrade (zero downtime), Java SLP security findings (2 batches), Oracle MySQL Critical Patches (CPUJAN2024, CPUJUL2024)
- ▶ Validated **43 Risk Indicators (rskInd)** in scoring API — Consul mock mode and live AIP integration — with schema propagation through RaaSOrchestrator, Hadoop consumers, billing, and reporting pipelines
- ▶ Led AIP transaction count mismatch investigation across VDP, RaaS HDFS, and AIP; identified timing and retry-logic root causes; unblocked production release
- ▶ Validated Global Risk Accounts, GFL Rate Limit, AIP Scoring Consumer scalability redesign (Spark 3.4→3.5, 4-processor architecture), V1/V2 API changes, request deduplication, and **Java 17 migration**

Stack: Java 17, Python, Go, Scala, Apache Spark 3.4/3.5, Delta Lake, HDFS, Apache Kafka (VRM), Redis, Cassandra, MySQL, Consul, Vault, Humio, Splunk, Grafana, Prometheus, JMeter, Jenkins, Artifactory, VisaKP/HMAC, SSL/TLS

Enterprise Fraud Management — Decision Manager UI

Nov 2020 — Apr 2023

Sr. Software Development Engineer in Test across Merchant & Acquiring Processing and Visa Risk departments.

- ▶ **Built FME Dashboard automation from scratch using Java and Python**: full suite covering table rendering, pagination, filters, and UI-API validation; adopted by wider QA team
- ▶ Found **critical Processor Config API (PCFGAPI) data integrity bug**: PATCH calls silently accepted invalid processor attributes (e.g., invalid fields for Citicon, Barclays), risking production merchant configuration corruption; validated fix pre-production
- ▶ Identified and validated fixes for **30+ UI bugs**: browser compatibility (IE/Safari/Firefox), timezone mismatch in IMS Flag calculations, WCAG 2.1 accessibility failures (400% reflow), and localization issues across 5+ report types

Stack: Java, Python, Angular, React, JavaScript, TypeScript, Selenium WebDriver, WebdriverIO, REST Assured, Postman, Jenkins, JIRA, HTML5, CSS3

- Led testing for financial surveillance platforms (including Behaviour Analytics and Regulatory Data pipelines); implemented Page Object Model automation using Selenium WebDriver, Java, Selenium Grid, Cucumber, TestNG, and Maven
 - Executed parallel cross-browser testing and automated test execution via Linux shell scripts; evaluated CI/CD pipeline severity for build releases
- Stack:** Java, Python, MySQL, PostgreSQL, REST Assured, TestNG, Selenium WebDriver, Selenium Grid, Cucumber, Jenkins, Apache JMeter, Maven, Linux

- Developed chat functionality for IPM HR services platform using ReactJS, Node.js, and RESTful APIs; reduced page load time to under 2s for 10+ websites, boosting conversion by up to 37%
 - Built Selenium automated test scripts with cross-browser parallel testing via Selenium Grid and JUnit
- Stack:** Java, Python, ReactJS, Node.js, JavaScript, MongoDB, RESTful APIs, Selenium WebDriver, Selenium Grid, JUnit, HTML5, CSS3, Webpack

PATENT

Release Management Automation — *"Release Management as a Service"*

USPTO — Granted | Visa Inc.

Co-inventor: automated pipeline from RISKRM Jira ticket state transition → JSON deployment config → Jenkins CI/CD for automated NonProd deployments, BVT execution, and Jira status tracking. Reduced deployment time by **20%**, eliminated manual steps, established single-source-of-truth release management with automatic audit trail.

EDUCATION & CERTIFICATIONS

Post Graduate Program in AI & Machine Learning: Business Applications	Dec 2025 – Aug 2026
McCombs School of Business, UT Austin	
AI & Machine Learning Bootcamp	Dec 2024 – Nov 2025
California Institute of Technology (Caltech)	
AWS Certified Developer – Associate	July 2021
Amazon Web Services	
Machine Learning Specialization	Feb 2023
Stanford University (Online)	
Master of Science, Computer Science	May 2019
George Mason University Fairfax, VA	
Bachelor of Technology, Computer Science	May 2017
Mahatma Gandhi Institute of Technology Hyderabad, India	